

From “Trust Me” to “Show Me”: The Principles of Explainable Artificial Intelligence

Session 2 Reading — Artificial Cognition and Explainable AI (5-Day Intensive)

Abstract

Session 1 established why modern AI systems are often opaque and why that opacity matters. This reading equips non-technical professionals with the conceptual vocabulary needed to engage with the field that has emerged in response — Explainable Artificial Intelligence (XAI). It begins with a clinical thought experiment — two doctors offering the same diagnosis, one with reasons, one without — to motivate the question that organises the chapter: what kinds of explanation do we need, from whom, and for what purpose? The reading carefully distinguishes three terms commonly used interchangeably (explainability, interpretability, transparency); maps the two principal dimensions along which XAI methods are sorted (scope: global versus local; source: intrinsic versus post-hoc); and engages directly with the most contested question in the field — the so-called accuracy–interpretability trade-off. It surveys the three motivations driving regulatory and organisational uptake of XAI: technical (debugging and validation), social (trust, recourse, contestability), and regulatory (GDPR Article 22, the EU AI Act, and sectoral rules in credit, health, and automotive). It closes with a sober note on what XAI can and cannot deliver, previewing the technique-level work of Session 3.

Keywords: explainability; interpretability; accuracy–interpretability trade-off; right to explanation; algorithmic accountability

1. The two doctors

Imagine you have just received a serious diagnosis. You consult two specialists. The first walks you through her reasoning: the scan shows X, the bloodwork shows Y, your symptoms align with Z; given your age, family history, and presentation, condition A is most likely; she has ranked B and C as less probable alternatives; here is how each diagnosis would change your treatment options. The second specialist looks at his computer screen and tells you, with the same confidence, that you have condition A. Asked why, he replies: “Trust me. I have seen many cases.”

Which specialist do you prefer? Most people prefer the first — not because they distrust the second’s clinical experience, but because the first has provided a *handle* on the decision: something to question, to verify against their own intuitions, to cross-check with a second opinion, to remember and explain to family. The second has provided a verdict.

The same disparity sits at the heart of contemporary debates about artificial intelligence. Many of the AI systems now deployed in healthcare, lending, hiring, criminal justice, and public administration behave like the second specialist — often more accurate than the alternatives, but mute on the reasoning behind their conclusions. The field of **Explainable AI (XAI)** is the institutional response. Its central claim is that, in domains where decisions affect human lives, prediction without explanation is increasingly unacceptable — technically, socially, and legally.

This reading equips you with the conceptual machinery to navigate that response. It does not yet teach you the techniques (those come in Session 3); it teaches you what techniques are *for*, what they can deliver, and what they cannot.

2. Three words that aren't synonyms

Practitioners and policymakers routinely use *explainability*, *interpretability*, and *transparency* as though they were interchangeable. They are not. Untangling them is the first step toward asking sharp questions of any AI vendor.

Transparency is a property of a model's *structure*. A model is transparent if a person can directly inspect its internals and follow how it converts an input into an output. A linear regression with five coefficients, or a decision tree with twelve nodes, is transparent. A deep neural network with several hundred billion parameters is not, even if its source code is open.

Interpretability is harder to pin down. Doshi-Velez and Kim offer the most-cited definition: interpretability is “the ability to explain or to present in understandable terms to a human.”¹ Zachary Lipton's essay “The Mythos of Model Interpretability” argues that the term has been used so loosely in the literature that it papers over several different desiderata — *simulatability* (can a person mentally execute the model?), *decomposability* (do the inputs, parameters, and computations each have an intuitive meaning?), and *algorithmic transparency* (do we understand the learning algorithm itself?).² On Lipton's analysis, “interpretability” is not one property but a family of them.

Explainability is typically reserved for the practice of generating human-understandable accounts of a model's behaviour, particularly for models that are not themselves interpretable. A SHAP plot attached to a deep-learning model's prediction is an explanation; the model is not interpretable, but its decisions are explained.

The distinction matters because the field has fractured along these lines. Cynthia Rudin argues that, in high-stakes settings, we should use **interpretable** (transparent, intrinsically understandable) models rather than build **explanations** on top of opaque ones. She writes: “There exists a widespread belief that more complex models are more accurate, meaning that a

¹Finale Doshi-Velez and Been Kim, ‘Towards a Rigorous Science of Interpretable Machine Learning’ (2017) arXiv:1702.08608, 2.

²Zachary C. Lipton, ‘The Mythos of Model Interpretability’ (2018) 61(10) *Communications of the ACM* 36.

complicated black box is necessary for top predictive performance. However, this is often not true.”³ The conflation of explainability and interpretability has, in her view, allowed practitioners to keep using black boxes when they should be using glass ones.

Table 1. Three terms that are not synonyms.

Term	What it is	Concrete example	What it implies for the designer
Transparency	A property of the model's structure; can be inspected directly	Decision tree with twelve nodes; linear regression with five coefficients	Choose a model whose internals are humanly readable
Interpretability	The capacity to be understood in human terms — simulatability, decomposability, algorithmic clarity	A scorecard: “+5 if X, -3 if Y, +2 if Z”	Constrain model complexity; align features with familiar concepts
Explainability	The practice of generating human-comprehensible accounts of model behaviour	SHAP plot attached to a neural-network's prediction	Pair the model with a separate explanation method

A useful operational rule: when a vendor describes their product as “explainable,” ask which of these three properties they are claiming. The answer reveals what you are buying — and what you are not.

3. Global versus local: flashlight or floodlight?

Once we accept that AI systems can and should be made comprehensible, the next question is: comprehensible at what scope? Two scopes dominate.

A **global** explanation describes how a model behaves *in general*. Across all the credit applications a bank's model has scored, which features matter most? Across all the chest X-rays a hospital's classifier has reviewed, which image regions drive the model's confidence? A global explanation is a *floodlight* — it shows the lay of the land.

A **local** explanation describes how the model behaved on *this particular case*. Why was Maria's loan denied? Why did the model flag *this specific* X-ray as suspicious? Why was *this* résumé filtered out? A local explanation is a *flashlight* — it illuminates a single point.

The two are not interchangeable; they answer different questions and serve different audiences. A regulator auditing whether a credit-scoring system discriminates against protected groups

³Cynthia Rudin, ‘Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead’ (2019) 1(5) *Nature Machine Intelligence* 206, 206.

needs global statistics about how the model uses gender-correlated features across all applications. An applicant denied a loan needs a local explanation that tells her what about *her* application drove the denial and what she would need to change to qualify next time. A data scientist debugging the model needs both — global views to spot systemic issues, local views to investigate specific failures.

The European Union’s AI Act recognises this dual need. Article 86 grants any “affected person” subject to a high-risk AI decision the right to obtain “clear and meaningful explanations of the role of the AI system in the decision-making procedure and the main elements of the decision taken”⁴ — a local right. Article 13’s broader transparency obligations on providers of high-risk systems⁵ are oriented more toward global accountability. Both layers serve real purposes; an organisation that supplies only one is incomplete.

4. Post-hoc versus intrinsic: recipe or tasting note?

Cutting orthogonally across the global/local distinction is a second dimension that matters even more for how an organisation builds its XAI capacity.

An **intrinsic** explanation comes from the model itself. The model is structured so that, when you ask why it made a particular decision, the answer falls out of its mathematics. A short decision tree explains itself: the rules *are* the model. A linear regression’s coefficients explain themselves: the contribution of each variable is the coefficient times the value. Intrinsic explanation is the *recipe* — read the model, and you know how the dish was made.

A **post-hoc** explanation is generated separately, after the fact, by a second algorithm whose job is to approximate or describe the first. The model itself is opaque; the post-hoc method (LIME, SHAP, Grad-CAM, counterfactual generators) inspects the model’s behaviour and produces a story about it. Post-hoc explanation is the *tasting note* — somebody else samples the dish and tells you what flavours they noticed.

The recipe–tasting-note metaphor lights up a critical caveat: a tasting note may be wrong. Two competent sommeliers can disagree about the same wine; two post-hoc explanation methods can produce different — sometimes contradictory — accounts of the same model’s behaviour. Krishna and colleagues call this “the disagreement problem in explainable machine learning”⁶ and show empirically that leading post-hoc methods regularly disagree on which features matter most for the same prediction. Slack and colleagues demonstrate something stronger — that adversaries

⁴Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) [2024] OJ L 1689/1, art 86(1).

⁵ibid art 13.

⁶Satyapriya Krishna, Tessa Han, Alex Gu, Javin Pombra, Shahin Jabbari, Steven Wu and Himabindu Lakkaraju, ‘The Disagreement Problem in Explainable Machine Learning: A Practitioner’s Perspective’ (2022) arXiv:2202.01602.

can construct biased models whose LIME and SHAP explanations *systematically hide* the bias.⁷ The recipe is what you actually cooked; the tasting note is one person's reading of it.

Table 2. The 2×2 of XAI methods: scope $\times$ source.

	Intrinsic (the recipe)	Post-hoc (the tasting note)
Global (floodlight)	Reading a decision tree's structure; inspecting a generalised additive model's shape functions	Permutation feature importance; global SHAP summary; partial-dependence plots
Local (flashlight)	Tracing the decision path through a small tree to a specific leaf; reading a linear model's per-feature contributions for one instance	LIME explanation for one prediction; local SHAP waterfall; counterfactual for one applicant

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Figure 1. The 2×2 of XAI scope and source, with example methods. A two-by-two grid. The horizontal axis is labelled "Source of Explanation" with poles "Intrinsic" (left) and "Post-hoc" (right); the vertical axis is labelled "Scope" with poles "Global" (top) and "Local" (bottom). The four quadrants are populated with the example methods in Table 2. Annotations note the trade-offs: intrinsic methods are guaranteed faithful but constrain model choice; post-hoc methods preserve model freedom but may be unfaithful; global methods serve auditors and developers; local methods serve affected individuals seeking recourse. The figure communicates that no quadrant is universally best — the choice depends on the audience and the decision context.

Most XAI techniques you will meet sit cleanly in one of these four quadrants. Most XAI strategies in practice combine techniques from multiple quadrants — a global view for auditors, a local view for the affected person, an intrinsic baseline against which to validate the post-hoc explanations.

5. The accuracy–interpretability trade-off — myth or reality?

The most contested question in the field is whether interpretability comes at a cost in accuracy. The conventional view, repeated in countless industry presentations, holds that there is a continuous trade-off: as models become more accurate (deep neural networks, gradient-boosted ensembles), they become less interpretable; as they become more interpretable (linear models, short decision trees), they become less accurate. The implication is that high-stakes domains demanding accuracy must accept opacity as the price.

⁷Dylan Slack, Sophie Hilgard, Emily Jia, Sameer Singh and Himabindu Lakkaraju, 'Fooling LIME and SHAP: Adversarial Attacks on Post Hoc Explanation Methods' in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* (AIES 2020) 180.

Cynthia Rudin dismantles this view in the domains where it matters most. Reviewing the empirical literature on tabular data with meaningful features — the kind used in credit scoring, recidivism prediction, healthcare risk stratification, and many regulated applications — she finds that simple interpretable models are *often as accurate* as black-box alternatives. Her own work on recidivism prediction showed that a sparse two- or three-feature scoring system matched the accuracy of the proprietary COMPAS algorithm.⁸ She concludes: “It is a myth that there is necessarily a trade-off between accuracy and interpretability.”⁹

A striking example comes from a 2015 study by Rich Caruana and colleagues on pneumonia risk prediction. The team built two models on hospital data: a high-accuracy neural network and a more interpretable generalised additive model (GAM). The GAM made an alarming discovery — the data showed that patients with asthma had *lower* mortality risk from pneumonia than otherwise-similar patients without asthma. The clinical truth is the opposite: asthma compounds pneumonia. The artefact arose because asthmatic patients with pneumonia were routinely admitted directly to intensive care, receiving such aggressive treatment that their outcomes ended up better than less-acutely-treated patients. A model that learned the raw correlation would, if used to *triage* incoming patients, send asthmatics home — the worst possible decision.¹⁰ The neural network learned the same correlation. The crucial difference was that the GAM *displayed* it; the neural network did not. Without interpretability, the failure mode would have been invisible until patients died.

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Figure 2. The accuracy–interpretability spectrum, with Rudin’s annotation. A horizontal axis runs from “More interpretable” (left) to “More accurate” (right). The conventional view, plotted as a smooth descending curve from upper-left to lower-right, suggests a strict trade-off. Rudin’s annotation, plotted as a flat plateau across most of the axis for tabular high-stakes domains, suggests that interpretable models often match black-box accuracy. The two curves diverge at the right-hand end of the axis, where richly unstructured data (images, audio, raw text) genuinely favour deep models. Marginal callouts identify the Caruana et al. pneumonia case (a successful GAM that revealed a fatal artefact invisible to the neural network) and Rudin’s recidivism work (a 2–3 feature scorecard that matched COMPAS).

The Caruana case generalises into a strong claim about high-stakes domains: where the cost of a hidden failure mode is grave, interpretability is not a luxury; it is a safety feature. This does not mean black-box models have no place — they remain competitive and often superior on richly unstructured data. But for tabular high-stakes decisions, Rudin’s recommendation is increasingly the default among careful researchers: *try the interpretable model first*.

⁸Jiaming Zeng, Berk Ustun and Cynthia Rudin, ‘Interpretable Classification Models for Recidivism Prediction’ (2017) 180(3) *Journal of the Royal Statistical Society: Series A* 689.

⁹Rudin (n 3) 206.

¹⁰Rich Caruana, Yin Lou, Johannes Gehrke, Paul Koch, Marc Sturm and Noemie Elhadad, ‘Intelligible Models for HealthCare: Predicting Pneumonia Risk and Hospital 30-Day Readmission’ in *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (KDD ’15, ACM 2015) 1721.

6. Why XAI? The technical motivation

Three distinct sets of reasons drive organisations to invest in XAI. Conflating them is among the most common errors in AI governance, because each implies different success criteria.

The **technical** motivation is the oldest. Engineers use XAI tools to *debug* their own models. If a fraud-detection system is performing poorly on a particular subset of transactions, feature-importance plots can reveal whether the model has learned a sensible rule or an absurd one. The literature is full of cautionary tales. The most cited is an image classifier trained to distinguish huskies from wolves that achieved high accuracy by detecting *snow in the background* — wolves were photographed in snow, huskies in studios; the model never learned anything about the animals.¹¹ Section 3 of Session 1 introduced this as the shortcut-learning or “Clever Hans” failure mode; XAI techniques are its principal diagnostic.

XAI is also the main tool for detecting **data leakage** — the contamination of training data with information that would not be available at deployment. A model that predicts hospital readmission may quietly learn to detect a feature like “discharge instructions printed in Spanish” that correlates with hospital location, drifting away from the clinical signal it was supposed to learn. Permutation importance and SHAP analysis can surface such patterns. Likewise, **distribution shift** — the slow drift of deployment data away from the conditions of training — can be diagnosed by tracking whether the features the model relies on remain stable over time.

XAI does for ML models what a flight-data recorder does for an aircraft: it makes failure analysis possible. An organisation that deploys ML without XAI is operating without a recorder.

7. Why XAI? The social motivation

Beyond debugging, XAI serves social functions that no technical metric captures.

The first is **calibrated trust**. Tim Miller’s influential synthesis of decades of social-science research on explanation argues that humans evaluate explanations along social dimensions — coherence, contrastive structure, selectivity — and not just along accuracy.¹² When users trust a system in proportion to how well they understand it, miscalibration is dangerous in both directions. Excessive trust in an opaque system produces *automation bias*: people defer to model outputs they should have questioned. Insufficient trust produces *under-utilisation*: people ignore valid recommendations because they cannot tell whether to believe them. Good explanations, on Miller’s account, are the principal mechanism for bringing trust into calibration with reality.

The second is **recourse**. When a model denies someone a loan, a job interview, or parole, the affected person has a legitimate interest in knowing what they would have to change to obtain a

¹¹Marco Tulio Ribeiro, Sameer Singh and Carlos Guestrin, “Why Should I Trust You?": Explaining the Predictions of Any Classifier' in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (KDD '16, ACM 2016) 1135.

¹²Tim Miller, 'Explanation in Artificial Intelligence: Insights from the Social Sciences' (2019) 267 *Artificial Intelligence* 1.

different outcome. Wachter, Mittelstadt and Russell formalised this as the **counterfactual explanation** — “if your annual income had been £35,000 instead of £30,000, your loan would have been approved” — and argued that such explanations satisfy a legitimate purpose without requiring the disclosure of the model’s full internals.¹³ Counterfactuals operationalise the moral intuition that algorithmic decisions about people should be *actionable*, not merely correct.

The third is **contestability**. The legitimacy of a public or quasi-public decision depends in part on the affected person’s ability to challenge it. A decision that cannot be challenged cannot be legitimate. When AI systems mediate access to housing, credit, healthcare, and welfare, the absence of meaningful explanation forecloses contestation and corrodes democratic accountability.

The Apple Card case (2019) illustrates the social stakes vividly. Multiple users — including the technologist David Heinemeier Hansson and Apple co-founder Steve Wozniak — reported that the algorithm-driven credit card offered women in their households substantially lower credit limits than men, even when finances were jointly held. Goldman Sachs, the issuing bank, responded that the algorithm did not use gender as an input.¹⁴ Whether or not that was technically true, the bank could not produce an explanation that satisfied affected users or regulators, and the New York Department of Financial Services opened an investigation. The reputational and regulatory cost was substantial, and the case became a textbook illustration of the social risks of unexplainable AI.

8. Why XAI? The regulatory motivation

The third motivation is the one that has moved fastest in the past decade and now drives much of the corporate investment in XAI.

8.1 GDPR Article 22 and the “right to explanation” debate

The European Union’s General Data Protection Regulation (GDPR), in force since 2018, established Article 22: “The data subject shall have the right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects concerning him or her or similarly significantly affects him or her.”¹⁵ Recital 71, which accompanies the GDPR but is not itself binding, mentions a “right to obtain an explanation of the decision reached after such assessment.”

¹³Sandra Wachter, Brent Mittelstadt and Chris Russell, ‘Counterfactual Explanations without Opening the Black Box: Automated Decisions and the GDPR’ (2018) 31(2) *Harvard Journal of Law and Technology* 841.

¹⁴Taylor Telford, ‘Apple Card Algorithm Sparks Gender Bias Allegations against Goldman Sachs’, *Washington Post* (Washington DC, 11 November 2019); the New York Department of Financial Services subsequently investigated and, in March 2021, did not find Goldman Sachs in violation of state fair-lending law on the evidence available, while noting that consumer-facing transparency had been inadequate. See NY DFS, *Report on Apple Card Investigation* (March 2021).

¹⁵Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data (General Data Protection Regulation) [2016] OJ L 119/1, art 22(1).

A vigorous scholarly debate followed. Wachter, Mittelstadt and Floridi argued that the binding text of Article 22 does *not* establish a right to explanation in the strong sense often claimed; Recital 71's language is non-binding, and the substantive rights in Articles 13–15 are limited to “meaningful information about the logic involved” — closer to a right to be informed than a right to a case-specific reasoned decision.¹⁶ Selbst and Powles offered the counter-argument that the “meaningful information” requirement, read in context, does amount to a right to explanation strong enough to require operational compliance.¹⁷ Goodman and Flaxman, in an early and widely-cited paper, took the strong view that GDPR's text “effectively creates a ‘right to explanation,’ whereby a user can ask for an explanation of an algorithmic decision that was made about them.”¹⁸

For non-technical professionals, the practical lesson is less the academic debate than the operational reality: data protection authorities across the EU have increasingly *acted* as though Article 22 requires meaningful explanations, and most large data controllers now design their systems on that assumption.

8.2 The EU AI Act

The EU AI Act (Regulation 2024/1689), which entered into force on 1 August 2024 and becomes fully applicable on 2 August 2026, makes the regulatory picture considerably more explicit. It classifies AI systems into four risk tiers: *prohibited* (e.g., social scoring by public authorities), *high-risk* (e.g., AI used in employment, credit, education, law enforcement), *limited risk* (e.g., chatbots), and *minimal risk*. For high-risk systems, the obligations are substantial.

Article 13 requires that high-risk systems be “designed and developed in such a way as to ensure that their operation is sufficiently transparent to enable deployers to interpret a system's output and use it appropriately.”¹⁹ Article 14 mandates effective human oversight — operators must be able to “correctly interpret the high-risk AI system's output.”²⁰ Article 86 introduces a clear right to explanation for affected persons: “Any affected person ... shall have the right to obtain from the deployer clear and meaningful explanations of the role of the AI system in the decision-making procedure and the main elements of the decision taken.”²¹

Article 99 sets penalties. Violating the Act's prohibitions on unacceptable-risk practices can attract administrative fines “of up to 35 000 000 EUR or, if the offender is an undertaking, up to 7 % of

¹⁶Sandra Wachter, Brent Mittelstadt and Luciano Floridi, ‘Why a Right to Explanation of Automated Decision-Making Does Not Exist in the General Data Protection Regulation’ (2017) 7(2) *International Data Privacy Law* 76.

¹⁷Andrew D. Selbst and Julia Powles, ‘Meaningful Information and the Right to Explanation’ (2017) 7(4) *International Data Privacy Law* 233.

¹⁸Bryce Goodman and Seth Flaxman, ‘European Union Regulations on Algorithmic Decision-Making and a “Right to Explanation”’ (2017) 38(3) *AI Magazine* 50, 50.

¹⁹AI Act (n 4) art 13(1).

²⁰*ibid* art 14(4)(c).

²¹*ibid* art 86(1).

its total worldwide annual turnover for the preceding financial year, whichever is higher”; breaches of high-risk-system obligations carry a lower ceiling of €15 million or 3 %.²² These figures are an order of magnitude beyond GDPR penalties at the prohibition tier.

8.3 Sector-specific obligations

Beyond the cross-cutting EU framework, sectoral regulations have long required explanation in specific domains.

In US credit, the **Equal Credit Opportunity Act** and its implementing **Regulation B**, alongside the **Fair Credit Reporting Act**, require lenders to provide “adverse action notices” giving a “statement of specific reasons” when an application is denied. A boilerplate “you did not meet our credit standards” is not compliant; the lender must provide a specific reason, such as “income too low” or “too many recent credit inquiries.”²³ When such denials are generated by ML models, the lender must produce a *faithful* reason — exactly the use case for local post-hoc methods like SHAP and LIME, or for intrinsically interpretable scorecards.

In healthcare, **HIPAA** does not directly require explanation, but the FDA’s evolving regulatory posture on AI-enabled medical devices increasingly emphasises interpretability as a condition for approval, particularly for systems used in clinical decision support.

In automotive, **ISO 26262** (functional safety) and **ISO/PAS 21448** (Safety of the Intended Functionality, SOTIF) require demonstrable safety arguments for AI components, which in practice require interpretability sufficient to support hazard analysis and certification.

Table 3. Three motivations for XAI, mapped to stakeholders, regulations, and success criteria.

Motivation	Primary stakeholder	Example regulations / standards	What “good” looks like
Technical (debugging, validation)	Engineers, data scientists, model-risk officers	Internal ML lifecycle standards; US Federal Reserve SR 11-7 (model risk management)	Failure modes identified before deployment; spurious correlations and leakage flagged
Social (trust, recourse, contestability)	Affected individuals; communities; civil society; journalists	GDPR Articles 13–15, 22; EU AI Act Article 86	Actionable counterfactuals; visible appeal routes; disaggregated harm statistics
Regulatory (compliance, audit)	Boards; compliance officers; auditors; regulators	EU AI Act (Arts 13, 14, 86, 99); GDPR Art 22; ECOA/Reg B; FCRA; HIPAA; ISO 26262; ISO/PAS 21448	Conformity documentation; demonstrable accountability; defensible deployment record

²²ibid art 99(3); art 99(4) sets the lower 15 M EUR / 3% tier for breaches of obligations applicable to high-risk systems.

²³Equal Credit Opportunity Act, 15 USC § 1691(d); Regulation B, 12 CFR § 1002.9; Fair Credit Reporting Act adverse-action requirements, 15 USC § 1681m.

9. A map of the XAI landscape

With the conceptual machinery established, we can now read the field's taxonomy more cleanly. Barredo Arrieta and colleagues' 2020 review, complemented and extended by Mersha and colleagues' 2024 update, organises XAI methods along several axes already encountered.²⁴

- **Model-specific vs. model-agnostic.** Some methods (decision-tree path tracing, attention visualisation in transformers) work only on certain model classes. Others (LIME, SHAP, permutation importance, counterfactual generators) work on any model that can be queried as a function.
- **Intrinsic vs. post-hoc** (Section 4 above).
- **Global vs. local** (Section 3 above).
- **Attribution-based vs. example-based.** Attribution methods (SHAP, LIME, Grad-CAM) explain a decision by assigning importance scores to input features. Example-based methods (prototypes, counterfactuals, influential training examples) explain by pointing at other data points — “this prediction is like these training cases” or “this is what would have to change.”

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Figure 3. A taxonomy tree of XAI methods. A horizontal tree diagram. The root node, “XAI methods,” branches first into “Model-specific” and “Model-agnostic.” Each branch then divides into “Intrinsic” and “Post-hoc”; each of those further into “Global” and “Local”; and finally each leaf is labelled with characteristic techniques: decision tree / GAM / sparse linear model (intrinsic, global, model-specific); decision-path / coefficient-reading (intrinsic, local, model-specific); permutation importance / partial-dependence / global SHAP (post-hoc, global, model-agnostic); LIME / local SHAP / counterfactuals / Grad-CAM (post-hoc, local). The figure is adapted from the taxonomy of Barredo Arrieta and colleagues (2020).

Christoph Molnar's textbook *Interpretable Machine Learning*, freely available online and updated to a third edition in 2025, organises the techniques in approximately this scheme and is the standard pedagogical reference for technical practitioners; Lavelle-Hill and colleagues' 2025 handbook in *Psychological Methods* does the equivalent for behavioural scientists.²⁵ You will meet specific techniques — LIME, SHAP, Grad-CAM, counterfactuals — in detail in Session 3.

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Figure 4. Choosing an explanation strategy: a decision flowchart. A flowchart in five decision nodes,

²⁴Alejandro Barredo Arrieta and others, 'Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI' (2020) 58 *Information Fusion* 82; Melkamu Mersha, Khang Lam, Joseph Wood, Ali K. AlShami and Jugal Kalita, 'Explainable Artificial Intelligence: A Survey of Needs, Techniques, Applications, and Future Direction' (2024) 599 *Neurocomputing* 128111.

²⁵Christoph Molnar, *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable* (3rd edn, self-published 2025), available at <https://christophm.github.io/interpretable-ml-book/>; Rosa Lavelle-Hill, Gavin Smith, Hanne Deininger and Kou Murayama, 'An Explainable Artificial Intelligence Handbook for Psychologists: Methods, Opportunities, and Challenges' (2025) *Psychological Methods* (advance publication, doi:10.1037/met0000772).

drawn top-to-bottom. (1) "Is the decision high-stakes (legal, medical, financial, criminal)?" Yes → consider intrinsically interpretable models first (Rudin route); No → either route acceptable. (2) "Is the input data tabular with meaningful features?" Yes → interpretable models are often competitive; No (images, audio, raw text) → black box with post-hoc explanation is often necessary. (3) "Who is the primary audience for explanations?" Affected individual → local + counterfactual; Regulator/auditor → global + documentation; Developer → both. (4) "Is recourse required by law (ECOA, AI Act Art 86)?" Yes → counterfactual or specific-reason local explanation. (5) "Have you validated explanations against ground truth or alternative methods?" If no, build that validation step before deployment. Arrows lead to four recommended strategies: "Interpretable-by-design," "Post-hoc local with audit," "Post-hoc global for governance," and "Combined approach."

10. What XAI can and cannot deliver

A reading on the principles of XAI would be incomplete without a sober note on the field's limits.

First, **post-hoc explanations can be wrong**. Krishna and colleagues' work on the disagreement problem, introduced in Section 4, shows that mainstream methods regularly produce inconsistent accounts of the same prediction; Slack and colleagues show that adversaries can deliberately construct models whose explanations *conceal* bias.²⁶ An explanation that looks reasonable is not necessarily faithful to the underlying computation.

Second, **explanation is not understanding**. Bhatt and colleagues surveyed practitioners deploying explainable ML in industry and found that explanations are most often consumed by *engineers and data scientists*, not by the affected end-users they nominally serve.²⁷ An explanation interface satisfying a regulator's audit may be incomprehensible to the loan applicant whose right it ostensibly enshrines. Building explanations that actually inform their nominal audience is, on the evidence, harder than the literature suggests.

Third, **explanations can manipulate**. A well-crafted explanation creates the *feeling* of comprehension and the willingness to comply. Where the underlying decision is wrong or unfair, a slick explanation may produce harm more effectively than no explanation at all. The literature on automation bias is sobering on this point.

Fourth, **the most important question may not be technical**. Whether a system should be deployed at all — whether an algorithm should mediate access to bail, healthcare, or employment — is a normative question that no amount of explanation resolves. As Taylor and Taylor remind us, treating AI cognitively is necessary precisely because the cognitive science of explanation is incomplete and the social science of accountability is unfinished.²⁸ XAI is necessary; it is not sufficient.

²⁶Krishna and others (n 6); Slack and others (n 7).

²⁷Umang Bhatt and others, 'Explainable Machine Learning in Deployment' in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (FAccT 2020, ACM) 648.

²⁸J. Eric T. Taylor and Graham W. Taylor, 'Artificial Cognition: How Experimental Psychology Can Help Generate Explainable Artificial Intelligence' (2021) 28(2) *Psychonomic Bulletin & Review* 454.

11. Looking ahead

You now have the vocabulary to engage with the techniques. Session 3 introduces the four most-used in practice: LIME, SHAP, attention/saliency methods, and counterfactual explanations. Session 4 applies them in sectoral case studies — healthcare, finance, manufacturing, autonomous systems, the public sector. Session 5 considers governance, ethics, and the integrative project.

If Session 1 asked “what kind of thing is an AI system?”, this reading has asked “what kind of explanation can we ask of it?” The remaining sessions ask the natural next question: “how, and to whom, do we deliver one?”

If you remember one thing from this reading, let it be the distinction between *interpretability* and *explanation*. An interpretable model carries its reasons on its sleeve; an explained model has a story attached. Stories can be helpful, true, and necessary. They can also be wrong, manipulated, or simply beside the point. The next three sessions will help you tell the difference.

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